

SD41 502A AMAZE



Engineering Journal
High Stakes 2024-2025

Innovate Award Submission Information Form

Only for events that are offering the Innovate Award

Instructions: Please fill out all information, printing clearly. This form should be included as per the instructions given in the Guide to Judging. Teams may only submit one aspect of their design to be considered for this award at each event. Submission of multiple aspects will nullify the team's consideration for this award.

Full Team Number: 502A

Brief description of the novel aspect of the team's design:

For effective scoring on neutral wall stakes, our rotational arm is equipped with a basket, pistons, and linear slides. Together these components allow our robot's arm to extend to various lengths.

Long Arm Advantages: Our robot can score on wall stakes during the autonomous period without crossing the autonomous line. It can also score the wall stake if game elements our other robots are in the way.

Short Arm Advantages: The robot can score more accurately, since the ring is more snug due to the anti-slip mat located on the triangle-supported L-channel. By having a shorter arm, it also allows our robot to utilise an aligner for consistent positioning.

Identify the page numbers and/or the section(s) where documentation of the development of this aspect can be found:

DO THIS ONCE NOTEBOOK IS COMPLETE

Explain why your submission is unique from other approaches to the problem it solves or task it performs:

This mechanism is unique from the other approaches to scoring on the neutral wall stakes as we are the only team to implement it into our design. It solves the disadvantages of the basic rotational arm by allowing variance in the length of the arm.

502A's Engineering Journal

High-Stakes

2024-2025

This digital notebook will clearly document our step-by-step design and development process.

Our approach during each practice involves:

1. Creating digital entries of important activities and accomplishments, marked with a date and time, to ensure evidence of chronological progression
2. Regularly reviewing and reflecting on past entries to identify areas for improvement and track progress over time.

Table of contents

This journal will be divided into sections; pages will be labeled here.

Chapter 1: The Team

Pg. 1 - 3 Team Members

Pg. 4 Team Member Assignments

Chapter 2: VEX High Stakes

Pg. 5 What is VEX High-Stakes 2024?

Pg. 6 - 9 High Stakes Game Overview

Pg. 10 - 12 High Stakes Game Formats

Pg. 13 - 15 High Stakes Scoring

Pg. 16 - 17 High Stakes Scoring: Rings

Pg. 18 High Stakes Scoring: The Ladder

Pg. 19 - 20 High Stakes Scoring: The Corners

Pg. 21 - 22 Understanding Game Rules

Table of contents

This journal will be divided into sections; pages will be labeled here.

Chapter 3: Robot - Acheron

Pg. 23 Robot Introduction: Acheron

Pg. 24-25 Drive Train

Pg. 26 Progress

Pg. 27 - 28 Characteristics We Want On Our Robot

Pg. 29 - 32 Types of Drive Bases to Consider Building

Explanations vary per page; depending on the content.

How to Read the Notebook

The title of the page will be in size 25 font

“Title”

Regular words and information in the notebook will be in font size ranging from 12-16.

“Normal”

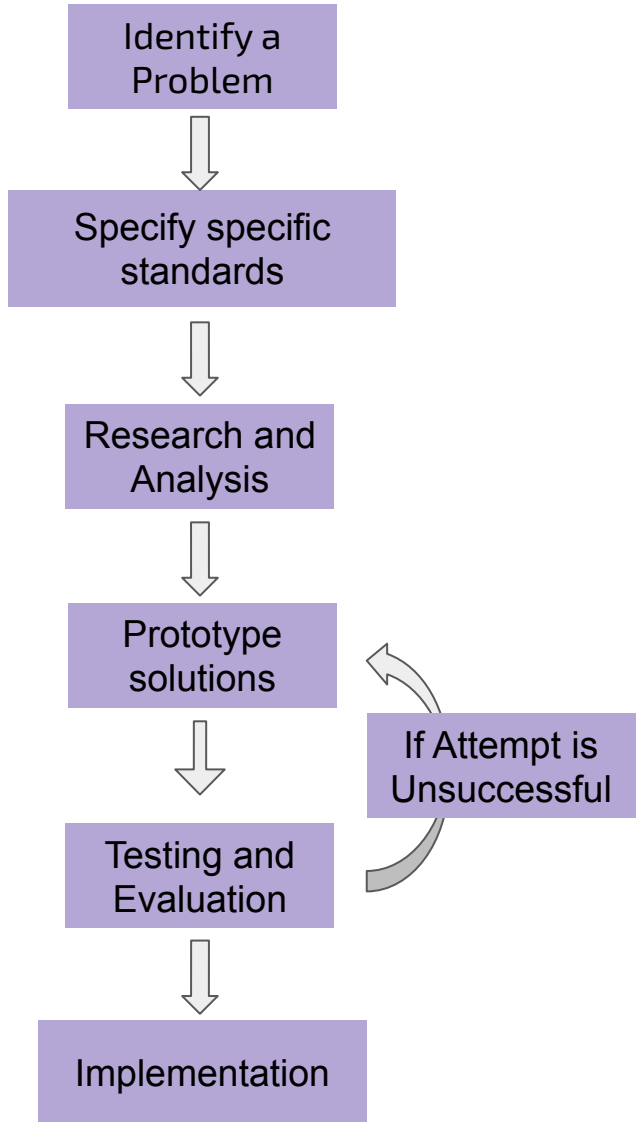
The bottom of all pages will contain the person who made the entry, the team name, and the page number. Font size 10, lavender color

“Name|Date|Page #”

Important information in the notebook will be highlighted or underlined

“IMPORTANT”

Development Process





Identify a Problem

- Describe the problem with words or images
- Be specific with the problem



Specify Specific Standards

- Specify any limitations or restrictions, materials, time
- Specify functional and quality requirements
- Develop a project timeline that includes for each task



Research and Analysis

- Explore various existing solutions
- Create/share ideas, combine concepts, and build upon each other's thoughts
- Determine if each solution is practical and achievable within the constraints of time and resources



Prototype Solutions

- Select the most promising solution
- 3D model the solution on Onshape
- Create the solution



Testing and Evaluation

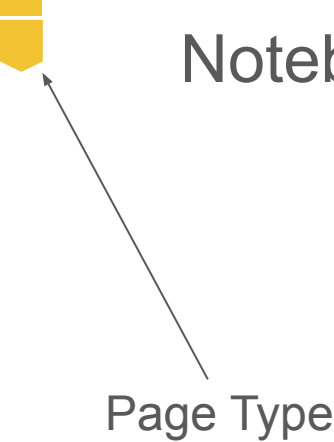
- Create an outline on how to test solution
- Conduct needed tests
- Create Conclusion based on results



Implementation

- If the results meet our expectations, integrate the solution onto our robot.
- If the results are not satisfactory, revisit step 3 to develop alternative solutions.

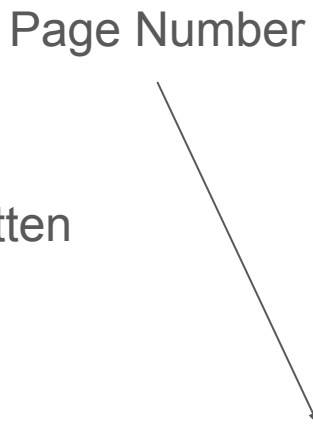
Notebook formatting



Page Type

A yellow tab icon is located at the top left. An arrow points from this tab to the text 'Page Type'.

Page Number



An arrow points from the text 'Page Number' down to the footer area.

Page Author



An arrow points from the text 'Page Author' down to the footer area.

Date Written



An arrow points from the text 'Date Written' down to the footer area.

Colors and Tabs

This notebook uses a color-coordinated page system:

- Each color scheme corresponds to a specific phase of the design cycle.
- For pages that cover multiple phases, the color is determined by the phase that represents the majority of the content.

This system ensures the notebook is well-organized and easy to navigate.

On the previous page, we included markings to indicate parts of the design process. To maintain consistency, these markings will now appear on the sidelines of the page and align with the color-coded system.

Different components of the game will have distinct color codes. We define four main parts of robotics, each marked with a specific color.

The start of each new section will have a small sticky note, making it easy to flip to the relevant section.

For example, game analysis content will be marked with turquoise sticky notes.

- **Turquoise for game analysis**
- **Green for planning/strategizing**
- **Dark purple for building**
- **Pink for coding**



Chapter 1

The Team

Meet the Team

502A consists of eight members, including some new to robotics and a few seasoned veterans. Together, we are ready to tackle the challenges of High Stakes, using our diverse skill set and experiences to achieve our best results.

Wellington Chan

Grade: 12

Experience: 3 years

Roles: Co-Leader, Driver,
Strategist, Builder, 3D
modeling/rendering



Arek Hong

Grade: 11

Experience: 3 years

Roles: Co-Leader, Robot Designer,
Builder, Strategist, Drive Team



Lucas Lu

Grade: 11

Experience: 3 years

Roles: Notebooker,
Videographer, Strategist



Siddarth S. Karthik

Grade: 10

Experience: 3 years

Roles: Planning, Coordinating,
Programming, Drive Team



Aiden Tong

Grade: 11

Experience: 1 years

Roles: Programming



Cannon Rowe

Grade: 11

Experience: 1 years

Roles: Builder



Ishan Kapadia

Grade: 11

Experience: 1 years

Roles: Builder, Backup Drive Team



Rachel Shin

Grade: 10

Experience: 1 years

Roles: Notebook Designer



Team Member Assignments

Task Assignments:

- **Arek:** Analyze game components.
- **Siddarth:** Review regulations surrounding the construction of the robot.
- **Wellington:** Examine general game rules
- **Lucas:** Analyze the structure of the tournament
- **Ishan and Cannon:** Analyze the structure of robot skills matches
- **Rachel:** Analyze the regulations regarding scoring



Chapter 2



VEX High Stakes

What is VEX High-Stakes 2024?

Problem Statement:

We are unfamiliar with the VEX competition rules. Our goal is to **thoroughly understand** them through detailed analysis.

Plan: Spend the first 3 days of the season analyzing the rulebook

Goal: Gain a thorough understanding of how High-Stakes works

Importance: Carefully examine the rulebook and other sources to **avoid unnecessary time-wasting** in the future.

Outcome: Ensure everyone has a **clear overview** of what needs to be done and a clear path forward.

High Stakes Game Overview

Game Components

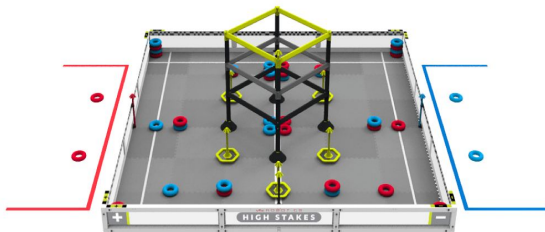
General Summary

The game is played on a 12'x12' field containing stakes, rings, the ladder positive/negative scoring corners.

The VEX competition field is a 12' x 12' square with an interior playing surface called the "Floor," composed of thirty-six grey foam field tiles arranged in a 6 x 6 grid. This Floor is enclosed by the "Field Perimeter," constructed from twelve straight sections.



The field is a 12'x12' square with "The Ladder" in the center. There are a total of 24 blue and 24 red rings. 5 mobile goals are positioned on the field. The field is also divided into four corners, two positive and two negative, demarcated by tape lines.



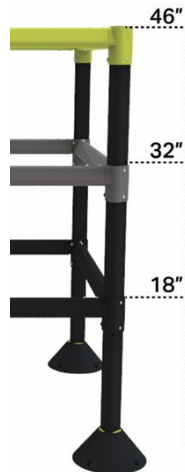
High Stakes Game Overview

The Ladder

One "Ladder" **centrally positioned** on the field, measuring 36" x 36" x 46".

Four vertical posts connected by three sets of horizontal rungs, establishing **three distinct climbing levels**. Robots can hang on the rungs.

- Black for the lowest
- Gray for the middle
- Yellow for the highest



The High Stake

One high stake located atop the vertical post **closest to the audience**

Colored yellow
Can only hold **one** ring.



High Stakes Game Overview

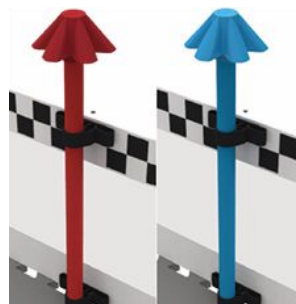
Mobile Goal

- Total of 5 hexagonal mobile goals
- Diameter of 10" and a height of 14.5"
- Serve as manipulatable mobile scoring targets for rings
 - Rings are scored on the stake in the center of the mobile goal
 - Can hold a maximum of 6 rings



Alliance Wall Stake

- Total of two alliance wall stakes (red and blue) positioned on each side of the field walls, parallel to the alliance stations
- Hold a maximum of 2 rings
- Height of 16.72"
- Robots are prohibited from interacting with their opponents' wall stakes



Neutral Wall Stakes

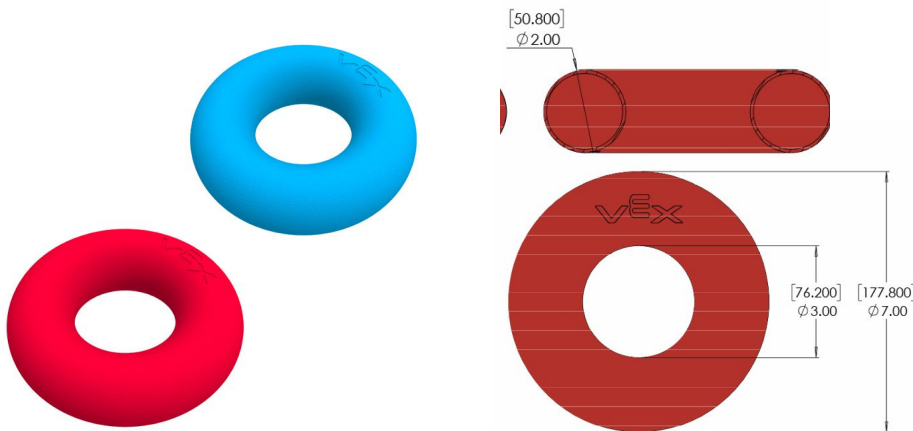
- Total of two neutral wall stakes positioned on each side of the field walls, perpendicular to alliance stations
- Hold a maximum of 6 rings
- Height of 25.09"
- Both teams can utilize any wall stakes



High Stakes Game Overview

Rings

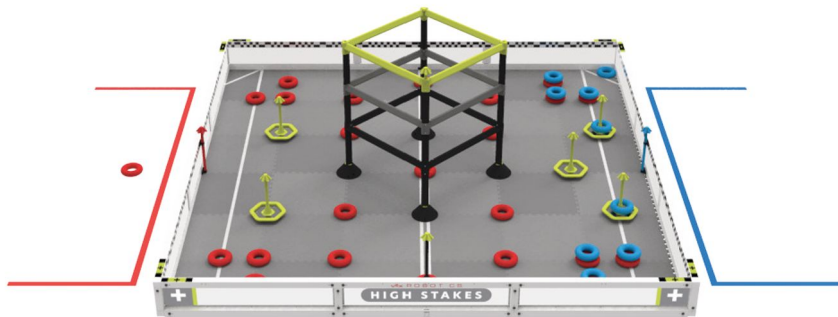
- Forty-eight rings are used per match
 - Forty four rings on the field at the beginning of each match
- Twenty-four rings are assigned to each Alliance—red and blue
- Four Alliance Rings are designated as Preloads.
 - Each alliance receives 2 preload rings in their team's color, 1 ring for each robot
- Weight: 0.25 lbs +/- 0.075 lbs
- Torus (donut) shape
 - Outer diameter of 7"
 - Inner "hole" diameter of 3"
 - Thickness of 2"



High Stakes Game Formats

Skills Competition (Robot Skills Matches)

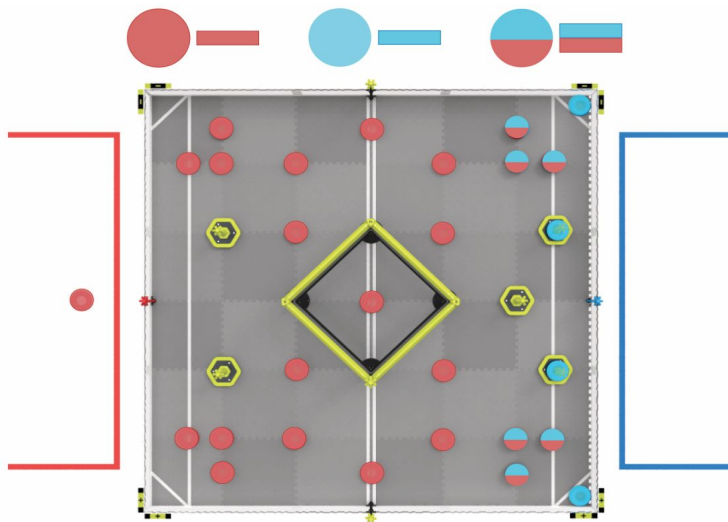
- Individual Competition: In robot skills matches, one robot competes alone to score as many points as possible
 - Driving Skills Matches (1 minute): Entirely driver-controlled
 - Autonomous Coding Skills Matches (1 minute): Robot operates entirely on pre-programmed instructions
- Teams ranked based on their combined scores from driving skills matches and autonomous coding skills Matches
- Must start the match in a legal starting position for the red Alliance
- All team members must stand in "Red Alliance Station"



High Stakes Game Formats

Skills Competition (Robot Skills Matches) Cont.

- Must start the match in a legal starting position for the red Alliance
- 23 **red rings** are present on field at the beginning
 - Only one red ring is a preload
- 10 **blue rings** are present on the field
- **Blue rings** can only be scored as top rings on stakes and are only worth points if
 - All 24 **red rings** have point values
 - No red rings can be above the blue ring
- No positive or negative corners
 - All mobile goals placed in corners will receive 5 points



Field Setup for skills: robot starts in the red section, total of 24 red rings and 10 blue rings on the field, each circle representing a ring of the corresponding color.

High Stakes Game Formats

Skills Competition (Robot Skills Matches)

- A total of 2 alliances, composed of 2 robots per alliance, red and blue compete against each other
- Objective: Score more points than the opposing alliance
- Points are earned by:
 - Scoring Rings on Stakes
 - Placing Mobile Goals in Corners
 - Climbing the Ladder at the end of the match
- Matches are decided into two periods:
 - Autonomous Period (15 seconds): Robot operate based on preprogrammed instructions and sensor input
 - Alliance with the most points receives the 6 points autonomous bonus
 - An Autonomous Win Point is awarded to alliances that completes the four assigned tasks during this period

Four tasks to achieve Autonomous Win Point

1. Three scored rings of the alliance's color
 2. Minimum of two stakes on the alliance's side
 3. Neither robot contacting/breaking the plane of the starting line
 4. One robot contacting the ladder
- Driver Controlled Period (1 minute 45 seconds): Robot is controlled remotely by the team

High Stakes Scoring

We will develop clarify scoring rules and create scoring strategies to prioritize tasks when planning and building our robot, ensuring effective game strategy.

Scoring Element	Description	Points
Autonomous Bonus	Awarded to the alliance with the most non-Climb, non-Corner points earned during the Autonomous Period	6
Scored Ring	Ring considered scored if it is "encircling" a stake	1
Top Ring	A ring is considered a "top ring" if it is the highest scored on a specific stake	3
Level 1 Climb	A robot's lowest point is above the foam tiles, but not higher than the black rung of the ladder	3
Level 2 Climb	A robot's lowest point is between the black rung and the gray rung of the ladder	6
Level 3 Climb	A robot's lowest point is between the gray rung and the yellow rung of the ladder	12
High Stake Bonus	Awarded to the alliance that has a ring scored on the High Stake at the end of the match. Each robot from that alliance that has climbed receives an additional 2 points	2

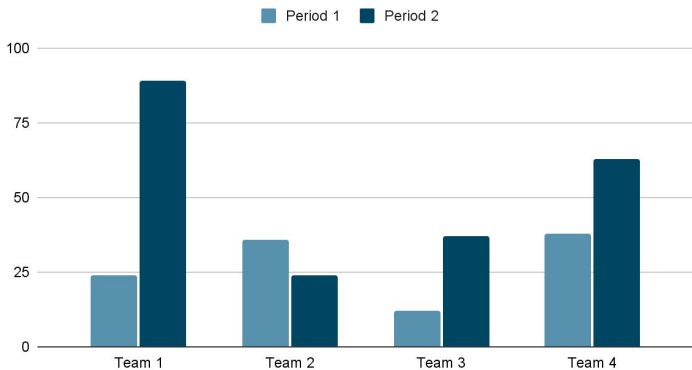
High Stakes Scoring

Scoring Element	Description	Points
Mobile Goal Placed in a Positive Corner	All scored rings on a mobile goal placed in a Positive Corner have their values doubled	2x
Mobile Goal Placed in a Negative Corner	Point values of all Scored Rings on mobile goal are reduced to zero and an equivalent number of points is subtracted from that Alliance's other Scored Rings	X0

Scores are calculated at the end of each match once all game elements are at rest.

The autonomous bonus is scored immediately after the autonomous period ends.

Points scored



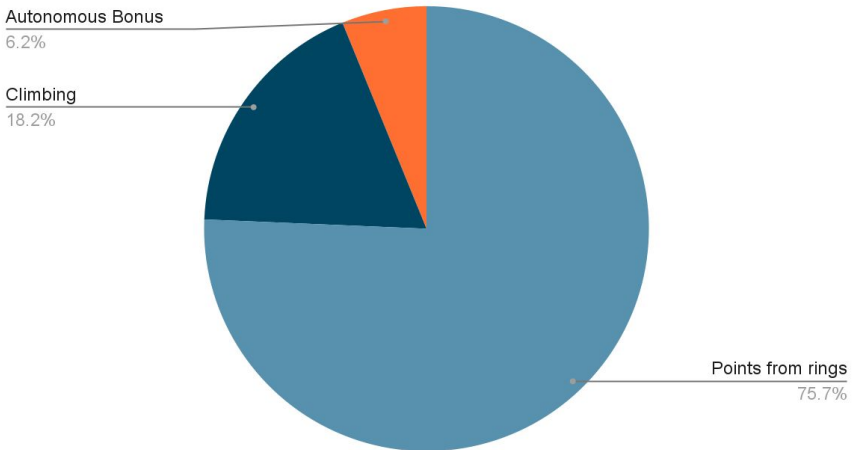
High Stakes Scoring

Total Points

- Around 50 points can be scored from scoring rings on stakes
 - Including top rings and double modifier
 - Including high stake
- 6 points from autonomous bonus
- 12 points from climbing (highest tier)

Can have a total of 68 points

Points scored



Observation

Rings scored on goals are the task worth the most points

Autonomous Bonus is worth the least amount

Endgame value varies on the design of the robot

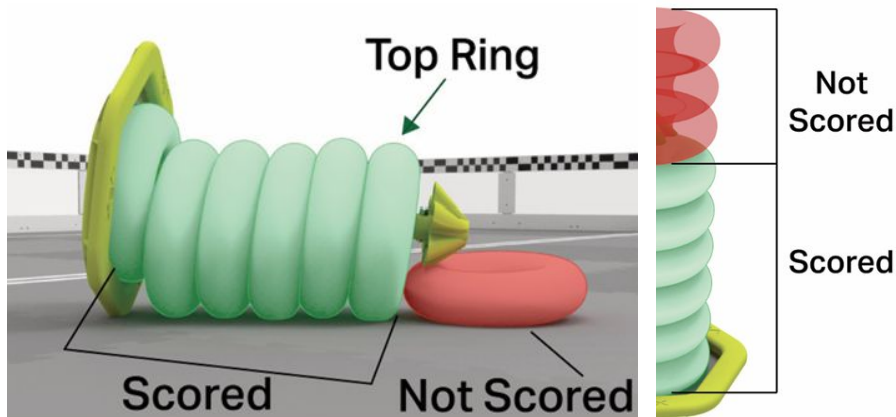
High Stakes Scoring: Rings

Ring is considered "Scored" on a Stake if it meets these criteria:

- The Ring is at least partially within the volume defined by the inner edges of the Ring
- The Stake does not exceed its total permitted number of Rings. Only 6 rings can be on a stake
- Mobile goals do not have to be upright

Ring is considered "Scored as a top ring" if it meets these criteria:

- Is Scored on a Stake (meets the criteria above)
- Is the furthest Scored Ring from the Stake's base
- If only one Ring is Scored on a Stake, it is still considered that Stake's Top Ring



High Stakes Scoring: Rings



Since there are a total of two rings that qualify as scored on this stake, the highest ring qualifies as the "Top Ring", earning 3 points instead of one



There are a total of five rings that qualify as scored on this stake, The highest ring (red ring) qualifies as the "Top Ring", earning 3 points for the red team. The four blue rings below the red ring are scored on the side of the blue team.

High Stakes Scoring: The Ladder

The ladder is the structure used for climbing.

A Robot **is considered** to be Climbing if:

- It has intentionally grasped, grabbed, or attached itself to any rungs on the ladder

A Robot is **not** considered to be Climbing if:

- It is touching these field elements:
 - Gray foam tiles
 - Mobile goals

Although the robot is hanging from the gray rung, it is touching the black rung; thus it qualifies for a Level 1 Climb

The robot meets all the criteria for a level 1 climb



High Stakes Scoring: The Corners

Summary: In head to head matches, Mobile Goals can be placed in one of four corners on the field, doubling or decreasing their point value. In skills, the above does not apply, placing a mobile goal in a corner earns an additional 5 points.

A mobile goal is considered placed in a corner if it touches the floor or white tape.

Corners do not affect *Climb* or *Autonomous Bonus Points*

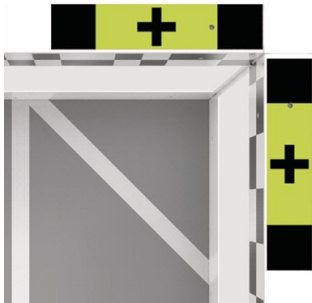
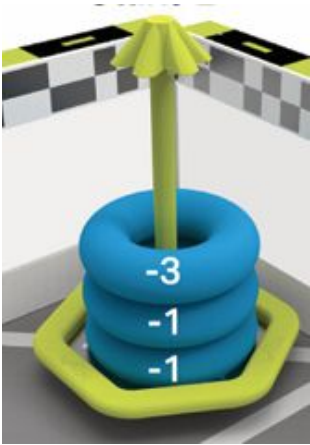
Positive Corner

- There are 2 positive corners on the field, designated by the "+" sign
- When a Mobile Goal is placed in a Positive Corner, the point value of all Scored Rings on that Mobile Goal is doubled
 - Scored Rings: Instead of 1 point each, they are worth 2 points each
 - Scored Top Rings: Instead of 3 points each, they are worth 6 points each

Negative Corner

- There are 2 negative corners on the field, designated by the "-" sign
- When a Mobile Goal is placed in a Negative Corner, the point value of all Scored Rings on that Mobile Goal becomes zero
- For every ring on that Mobile Goal, an equivalent number of points is deducted from the alliance's other Scored Rings
 - Scored Ring: Subtracts 1 point from your alliance's total score
 - Scored Top Ring: Subtracts 3 points from your alliance's total score

In this example, 3 rings are in a mobile goal placed in the negative corner. A total of 5 points will be taken away from the blue alliance



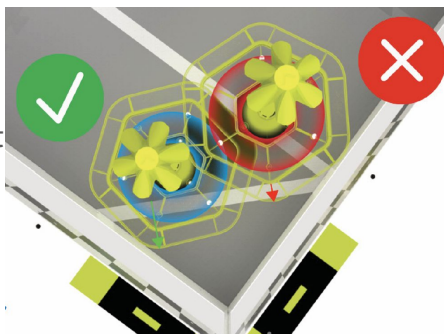
High Stakes Scoring: The Corners

When multiple Mobile Goals meet the requirements to be considered "Placed" in the same Corner, these tie breakers will determine which single mobile goal earns the points:

1. Mobile Goal with its base **extending farthest** into the corner wins
2. If the bases are too close, the Mobile Goal with the **straightest Stake** (perpendicular to the floor) wins
3. If both base and stake are too close, the Mobile Goal with its flexible top extending **farthest into the corner** wins

If it's still too close to call, neither Mobile Goal is considered placed

The base of the left Mobile Goal extends farther into the corner, it is considered scored.



Understanding Game Rules

Goal: We want to understand all the rules of "High Stakes" to prevent misunderstandings.

Starting Position

- <R4> Size Limits
- Robots must fit within an 18" x 18" x 18" volume at the start of each match

- <SG1> Starting a match:
- Robot must be positioned, so that it is contacting its alliance's starting line
 - Not touching any scoring objects other than one preload ring
 - Not contacting any other robots.

- <SG2> Horizontal Expansion
Robots can expand horizontally during the match, but:
- Can never exceed 24" x 18"
 - Can only expand horizontally in one direction

- <SG3> Vertical Expansion
Robots can expand vertically, but:
- Can never contact or "break the plane" of more than two planes of the Ladder simultaneously.

Understanding Game Rules

<SG9>

Interaction with Climbing Robots

- Intentional strategies to damage or tip opponents off the Ladder are not allowed, but incidental contact is acceptable

<SG6>

Possession Limits

- Robots cannot possess more than two rings or one Mobile Goal at once
- Rings on a Stake do not count towards the ring possession limit.

<SG7>

The Autonomous Line

- During the Autonomous Period, robots cannot touch any elements on the opposing alliance's side of the Autonomous Line

Progress

Throughout the months of September and October, we'll start our project, focusing on planning and familiarization with the game. This period will be dedicated to activities such as brainstorming sessions, gain a deeper understanding of the game's intricacies. By dedicating these two months to planning and exploration, we aim to lay a solid foundation for the next phases of our project.

In November, we'll complete our first robot. Depending on our performance, we'll carefully assess and implement any necessary changes to enhance our strategies and overall approach.

Moving into December, our focus will shift towards a detailed brainstorming session where we'll identify and execute essential modifications to both our robot and code. This phase is crucial for refining our technological components and ensuring optimal functionality.

In January, our primary emphasis will be on the physical construction of our robot. Simultaneously, we'll dedicate substantial efforts to coding, utilizing the LemLib template as a foundation.

As we transition into February, preparations for the upcoming provincials take center stage. Emphasizing both driving skills and autonomous functionalities becomes imperative during this period, laying the groundwork for a competitive edge in the imminent events.

March is the provincial competition, and our aspirations are high. We aim for good skills scores, very good driving capabilities, and the ambition to secure a spot in the world championships. We anticipate a successful representation on the global stage.



Chapter 3

Robot - Eiffel

Robot Introduction: Acheron



About Acheron

Acheron is a one of a kind robot. Though this robot may appear like a standard “Ladybrown” robot, this robot has a few tricks up its sleeve. From hidden game starting mechanisms to a unique arm, Acheron will surely AMAZE the competition.

In this chapter, we will be going through our full design process of Acheron.

Creating The Drivetrain

Problem Statement

We need to design a drive train that will allow us to maneuver the field in a smooth and stable manner.

Specifics

Drivetrain must have torque though if torque is the main priority, the drivetrain will lack speed. This is the same vice versa. Determining an appropriate gear ratio is the key to our robot's success.

The choice of motor cartridges in the drivetrain will matter. The higher the speed of the motor cartridge, the lower the torque it will have thus, leading to overheating or weak pushing power when in a defensive situation. This is also the same vice versa. Additionally, deciding the amount of motors that will be used in the drivetrain will have a significant impact.

Determining the type and size of the wheels will have great effect in maneuverability, agility and pushing power. Determining which wheels to in the drivetrain is vital.

Creating The Drivetrain

The Plan

Spend one week researching, brainstorming and designing an efficient drivetrain.

Goals

- Determine wheel types and sizes to be used in the drivetrain
- Research the most efficient gear ratios for the drivetrain
- Determine motor cartridges to be used in the drivetrain
- Determine the final motor count for each side of the drivetrain

Ideal Outcome

We hope to design and construct an effective drivetrain that balances speed and strength.



Plan for the next week

- **Day 1:** Research different drivetrain configurations (4, 6, 8-wheel drives) and their pros and cons.
- **Day 2:** Analyze specific needs for speed, stability, maneuverability, and field characteristics.
- **Day 3:** Sketch designs for each configuration, considering wheel types, motor placement, and control systems.
- **Day 4:** Evaluate feasibility, considering cost, build time, and complexity.
- **Day 5:** Choose the most suitable configuration based on evaluations.
- **Day 6:** Finalize design, selecting specific components like motors, wheels, and gears.
- **Day 7:** Begin prototyping or detailed planning for the final build.

Brainstorm Solutions - Drivetrain

Before we begin designing different types of drivetrains, we must determine which wheels to use

Omni-Directional Wheels:

- **Functionality:**
 - Allow movement in any direction (sideways, diagonal).
- **Usage:**
 - Common in holonomic drivetrains (X-drives, H-drives, Kiwi drives).
 - Often combined with traction wheels in tank drives.
- **Sizes:**
 - **4-inch wheels:**
 - Fast but prone to getting stuck on game objects.
 - Poor center of gravity.
 - **3.25-inch wheels:**
 - Balance of speed and strength.
 - **2.75-inch wheels:**
 - Great center of gravity.
 - Strong and won't climb over game objects.
- **Anti-Static:**
 - 2.75" anti-static omni wheels reduce spacing issues compared to older styles.



Brainstorm Solutions - Drivetrain

Traction Wheels:

- **Functionality:**
 - Provide grip and pushing power.
- **Usage:**
 - Used in tank drives for stability and to prevent drifting.
 - Often combined with omni-wheels for balanced performance.
- **Typical Combination:**
 - Typical setup: Two traction wheels sandwiched between two omni-wheels on each side.
- **Center Wheel:**
 - A center traction wheel can prevent drift in autonomous routines.



Brainstorm Solutions- Drivetrain

Before we begin designing different types of drivetrains, we must compare the three types of motors, their strengths and when and where to use them.

Blue Cartridge (600 RPM)

Speed: The blue cartridge is the fastest, with a speed of 600 RPM.

Torque: It has low torque, making it suitable for mechanisms that prioritize speed over power.

Usage: Typically used for mechanisms requiring extra speed.

Friction: Places little friction on the motor.

Green Cartridge (200 RPM)

Speed: The green cartridge has a speed of 200 RPM, which is considered moderate.

Torque: It provides increased torque compared to the blue cartridge, making it suitable for mechanisms that require a balance of speed and strength.

Usage: Usually used for mechanisms requiring a balance of strength and speed.

Friction: Loses speed to friction in the cartridge.

Red Cartridge (100 RPM)

Speed: The red cartridge is the slowest, with a speed of 100 RPM.

Torque: It has the highest torque among the three cartridges, making it suitable for mechanisms that require significant power.

Usage: Not explicitly stated in the sources, but implied to be for mechanisms that require very high torque and slow speeds.

- Cartridges can be geared to achieve specific speeds or torque.
- Example: A 600 RPM motor can be geared down to 300 RPM using a 36-tooth gear driving a 72-tooth gear.

Drivetrain - What do we want?

Drivetrain directly affects our robot's mobility, maneuverability, and performance during matches/skills

1. Field Navigation

The field contains many obstacles such as Mobile Goals, Stakes, and other robots.

A well-designed drivetrain enables our robot to move smoothly and quickly across the field, allowing it to:

- Collect Rings.
- Navigate to specific locations, such as Positive Corners for Mobile Goals.
- Climb structures like the Ladder for end-game points.
- Without an effective drivetrain, a robot would struggle to execute basic tasks, limiting its ability to score.

2. Speed vs. Torque Balance

A good drivetrain balances speed (how fast the robot moves) and torque (the robot's ability to push or carry objects).

In offensive strategies, a fast drivetrain allows the robot to reach scoring objects quickly or beat an opponent to a Mobile Goal.

In defensive strategies, a high-torque drivetrain can push opposing robots or resist being pushed, allowing for greater control of field elements.

Drivetrain - What do we want?

3. Agility for Precision Tasks

A responsive and agile drivetrain lets drivers maneuver the robot with accuracy, improving scoring efficiency during the Driver Controlled Period.

4. Interaction with Other Robots

A robust drivetrain allows a robot to withstand defensive maneuvers, avoid entanglements, and counter defensive strategies by moving fluidly around opponents. A drivetrain with good traction can help prevent our robot from being pushed away from scoring zones.

5. Strategy Execution

The drivetrain dictates how effectively we can implement a strategy. Whether the goal is to score points rapidly or play defense by blocking opponents, the drivetrain performance is pivotal. Poorly designed drivetrains can hinder a robot's effectiveness, while high-performance drivetrains give us a leverage..

Ideal Characteristics We Want On Our Robot:

Speed

Throughout the entire High Stakes game, speed plays a crucial role in achieving success on the playing field. With a limited timeframe, our robot needs to be able to complete various tasks, swiftly navigate and manipulate game elements to maximize its scoring potential. A faster robot gains a strategic advantage by effectively covering the field, swiftly collecting and depositing rings, and ultimately contributing the team's overall performance in this dynamic and fast-paced competition.

Resistance from being Pushed Around

- Maintaining Position
- Scoring Efficiently
- Being able to resist being pushed around is essential because it can affect your gameplay or the robot's movement

Agility

- Important for obstacle navigation and outmaneuvering opponents
- Being agile is essential. We will need to be able to quickly maneuver throughout the field.



Pushing Power

- Defense and Offense
- Pushing power is vital for effective navigation, game piece manipulation, and strategic dominance, offering advantages in matches by facilitating efficient movement and control over opponents.

Durability

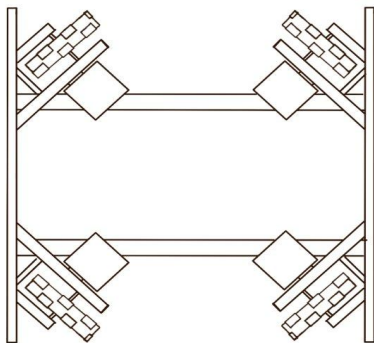
- Reliable and consistent
- Minimizes wear and tear
- A durable robot is essential to withstand challenges, collisions, and repeated movements, ensuring reliability and minimizing malfunction in during matches.

Scoring Mechanism

- Must have the ability to match load, not only in games, as well as in skills
- Best way to score points

Types of Drive Bases to Consider Building: XDrive

Aspect	Pros	Cons
Weight	Relatively light (no outer C-channels).	Lightness reduces structural integrity, making it less durable.
Speed	Can move quickly due to low torque.	Quick movements with less torque lead to instability.
Torque	-	Wheels have less torque (cannot be geared).
Traction	-	Little friction in every direction; poor traction with the ground.
Gearing Flexibility	-	Cannot gear the wheels to adjust speed or torque.



H Base

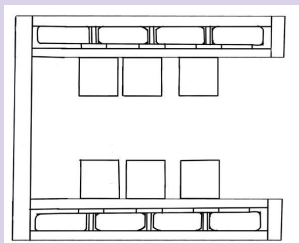
Aspect	Pros	Cons
Base Structure	Strong, efficient, and well-structured.	Middle wheel may be difficult to secure.
Wheel Support	Numerous wheels provide excellent support.	-
Customizability	Highly customizable; can use gears and motor cartridges to adjust speed/torque.	-
Resource Usage	-	Uses five motors, consuming more resources.

6-Wheel Drive Base

Aspect	Pros	Cons
Stability	Enhanced stability with 6 wheels, ensuring better control during games.	-
Traction & Versatility	Balanced traction and versatility for effective movement.	-
Customizability	- Gear ratio can be customized to optimize torque and speed. -Can use a mix of traction and omni wheels to fit specific needs.	
Maneuverability	Improved maneuverability for precise movements during competition.	-
Motor Usage	-	Requires 6 motors, consuming a significant portion of motor resources.
Size & Space	- Challenges in maintaining a compact design while ensuring stability.	- Large size may limit space for other components.

8-Wheel Drive Base

Aspect	Pros	Cons
Pushing Power	Effective pushing capabilities, suitable for game requirements.	-
Structural Integrity	Increased structural integrity, providing strong support during gameplay.	-
Customizability	Gear ratio can be customized to balance speed and torque.	-
Simplicity	Streamlined design with no need for extra parts or additional C-channels.	-
Adaptability	Versatile and adaptable to different game scenarios.	-
Space Constraints	-	Limited space for odometry wheels due to six motors in the middle.
Motor Usage	-	Requires six motors for sufficient power, leaving only two motors for additional functions.



Drivebase Analysis

Why X-Drive and Mecanum Drive Were Eliminated:

1. Lack of Adjustable Torque/Speed:

- **Issue:** X-drive and Mecanum drives cannot easily adjust torque or speed because gearing is difficult to implement.
- **Result:** Motors are locked at fixed speeds (200 RPM or 600 RPM), which limits flexibility during gameplay.

2. Fixed Speed Limitations:

- **Issue:** Fixed speeds (200 RPM or 600 RPM) cannot be adjusted to match specific game requirements.
- **Result:** This rigidity can cause problems in games where varying speeds and torque are needed for different tasks.

3. Complexity and Maintenance:

- **Issue:** Both X-drive and Mecanum drives are mechanically complex and harder to maintain.
- **Result:** This increases the risk of mechanical failures during competitions.

4. Poor Pushing Power:

- **Issue:** These drivetrains lack the traction and stability needed for effective pushing.
- **Result:** They struggle in tasks that require strong pushing power, such as moving game objects or defending.

Why the 8-Wheel, 6-Motor Drive is a Better Choice:

1. **Adjustable Torque/Speed:**
 - **Benefit:** The 8-wheel drive allows for customizable gear ratios, enabling us to balance speed and torque as needed.
 - **Result:** This flexibility ensures optimal performance for different game scenarios.
2. **Enhanced Stability and Traction:**
 - **Benefit:** The 8-wheel design provides superior stability and traction, making it ideal for pushing and precise movements.
 - **Result:** The robot can handle heavy tasks and maintain control on uneven surfaces.
3. **Efficient Motor Usage:**
 - **Benefit:** Using 6 motors (instead of 8) reduces resource consumption while still providing sufficient power.
 - **Result:** This leaves room for additional mechanisms and reduces strain on the robot's power system.
4. **Simpler Maintenance:**
 - **Benefit:** The 8-wheel drive is easier to maintain compared to X-drive or Mecanum drives.
 - **Result:** This reduces downtime and ensures reliability during competitions.

Comparing 6 and 8 Wheel drives

1. **Stability:**

- More contact points with the ground provide enhanced stability, reducing the risk of tipping or losing balance, especially during rapid movements or sharp turns.

2. **Maneuverability:**

- A better distribution of omni and traction wheels allows for smoother and more controlled movements, enabling complex maneuvers without compromising stability.

3. **Motor Usage:**

- Efficient motor configuration, potentially requiring fewer motors than the number of wheels, leading to better power usage and heat dissipation.

4. **Weight Distribution:**

- Even weight distribution prevents excessive load on any single wheel, reducing wear and tear and improving traction.

5. **Redundancy:**

- The likelihood of continued operation even if one wheel fails is higher, ensuring reliability during competition.

6. **Mechanical Complexity:**

- While more complex, the benefits in stability and performance justify the added complexity.

7. **Speed and Control:**

- Improved stability allows for faster speeds without losing control, especially with proper gearing and motor selection.

Importance of Different RPM Values

To better understand why 600 RPM is optimal, let's compare it to other potential RPM values:

1. Low RPM (e.g., 300 RPM)

Comparison

Characteristics:

High torque, allowing for greater force when moving or pushing objects.

Slower rotation speed, resulting in reduced linear speed.

Implications:

While high torque ensures stability and strong pushing power, the slower speed reduces the robot's ability to traverse the field quickly.

A low RPM might also make it harder to achieve the necessary velocity to outmaneuver opponents or reposition efficiently during matches.

2. Medium RPM (e.g., 600 RPM)

Characteristics:

Balanced torque and speed, enabling both agility and sufficient pushing power.

Suitable for rapid traversal of the field while maintaining control.

Implications:

This RPM aligns well with the 55 to 87 inches per second range recommended for a 6-motor drivetrain, ensuring efficient mobility.

With a gear ratio of 3:4, this RPM provides sufficient torque to overcome resistance from the field surface while maintaining a high enough speed to maximize scoring opportunities.

3. High RPM (e.g., 900 RPM)

Characteristics:

High speed, allowing for rapid movement across the field.

Reduced torque, which may compromise stability and pushing power.

Implications:

At excessively high RPMs, the drivetrain may struggle to maintain traction, leading to wheel slippage or mechanical failures.

Additionally, higher RPMs consume more power, potentially leaving insufficient energy for other critical functions like scoring mechanisms or climbing.

Determining the Gear Ratio

Performance Metrics

1. Linear Speed Calculation

Using the formula:

RPM: 600

Wheel Diameter: 4 inches

Wheel Circumference: inches

Gear Ratio: 3:4 (or 0.75)

Substituting the values:

This speed falls within the optimal range of 55-87 inches per second for a 6-motor drivetrain, ensuring efficiency without sacrificing control.

2. Torque Optimization

The 3:4 gear ratio increases torque, allowing the robot to:

Push Mobile Goals effectively

Overcome surface resistance

Maintain traction during sharp turns

3. Maneuverability

The omni-wheel setup enables holonomic movement, allowing:

Smooth strafing

Quick reorientation

Fast and precise field positioning

Determining the Gear Ratio

Addressing Potential Challenges

1. Turning Consistency

Issue: Wheels may "buffer" or move unpredictably during turns

Solutions:

Calibrating motor signals to ensure synchronized movement

Using independent wheel control for precise adjustments

2. Voltage Sag Management

Using 6 motors instead of 8 minimizes battery strain, reducing voltage sag and ensuring stable power distribution

3. Field Surface Resistance

The drivetrain's gear ratio and torque optimization ensure smooth operation on uneven surfaces

Why This Configuration is Ideal

1. Speed and Agility

600 RPM motors with a 3:4 gear ratio provide a linear speed of 83.8 in/s, allowing rapid field traversal

2. Holonomic Movement

The omni-wheel setup grants superior maneuverability, essential for navigating obstacles and precisely aligning with game elements

3. Power Efficiency

The 6-motor configuration reduces weight and power consumption, preserving energy for scoring mechanisms and climbing

4. Reliability and Control

Independent wheel control and calibration enhance movement consistency, reducing risks of drivetrain failure

Ideal Characteristics of a Scoring Mechanism

Accuracy

- Description : The scoring mechanism must consistently score the rings into the target (Stakes) from various distances and angles. Precision is critical, especially when Rings need to be placed on specific Stakes to maximize points.
- Importance : Missing shots can cost valuable points, so accuracy is a top priority.

Speed

Description : The faster our robot can score Rings, the more opportunities it will have to complete other tasks, such as placing Mobile Goals or climbing at the end of the match.

Key Metric : A high rate of scoring is desirable, but it should not come at the expense of accuracy.

Importance : Speed allows us to outpace opponents and maximize our scoring potential.

Power Efficiency

Description : The mechanism should be energy-efficient.

Importance : Efficient power usage allows our robot to allocate power to other critical functions, such as mobility and climbing, without compromising performance.

Compact Design

Description : The mechanism should be compact to conserve space on the robot.

Importance : A compact design allows room for other essential components, such as the drivetrain, sensors, and mobile goal handlers, ensuring that the robot remains balanced and functional.

Ease of Use

Description : The mechanism should be easy to operate, both in autonomous and driver-controlled modes.

Importance : Drivers should be able to execute shots with minimal effort, and the programming should be straightforward to implement. Ease of use would allow us to easily implement and utilize the mechanism at any time.

Robot Overview

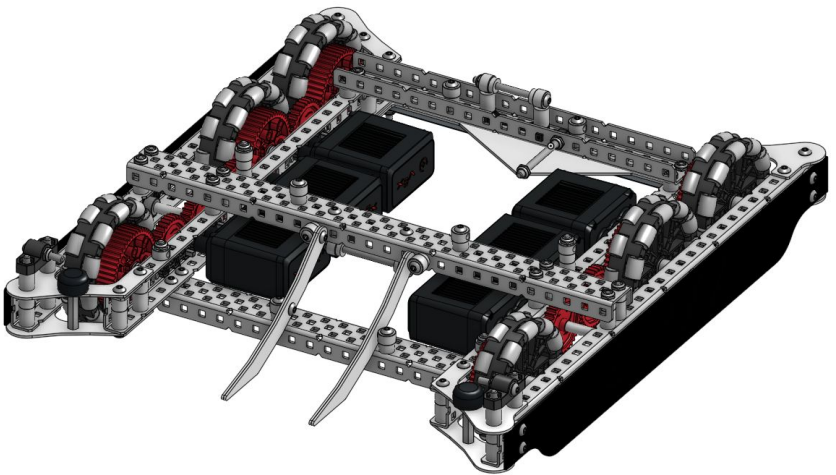
Our robot is meticulously engineered to excel in the VEX VRC "High Stakes" game, focusing on efficiency, speed, and versatility. The design incorporates a six-motor drivetrain, specialized intake and conveyor systems, a multifunctional doinker, a dynamic wall stake arm, a reliable climbing mechanism, a precision clamp, and advanced odometry wheels. Each component is optimized to enhance performance during both autonomous and driver-controlled phases.

Drivetrain

- **Motor Configuration:** The drivetrain utilizes six high-efficiency 66-watt motors configured in a 3:4 gear ratio, achieving a top speed of 450 RPM. This balance between speed and torque ensures rapid movement across the field while maintaining control during intricate maneuvers.
- **Wheel Selection:** Equipped with six 3.25-inch omni-directional wheels, the robot benefits from enhanced maneuverability, allowing for swift directional changes and smooth navigation around obstacles.
- **Weight Optimization:** To maximize acceleration and speed, we've minimized the robot's weight by securing gears to wheels using zip ties. This innovative approach reduces rotational inertia, enabling quicker starts and stops.

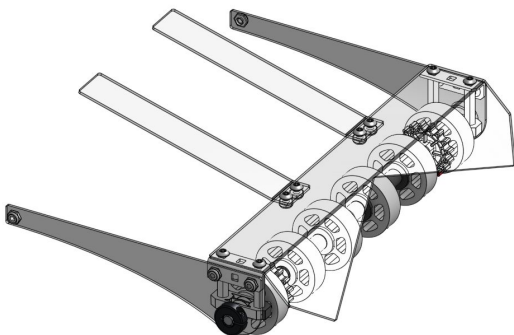
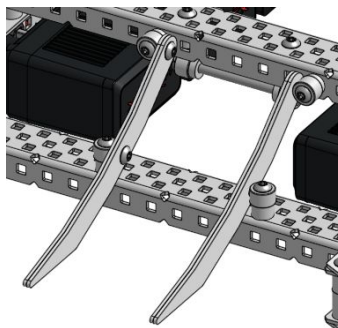
Dual Funnel System

- **Front Funnel:** Designed for versatile ring intake, the front funnel allows collection from various angles, increasing the likelihood of successful ring acquisition during dynamic play.
- **Rear Funnel:** The rear funnel is tailored for precise mobile goal (mogo) handling. It guides the mogo into the optimal position for secure clamping, streamlining the process during both autonomous and driver-controlled periods.



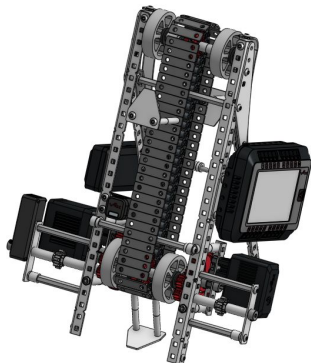
Intake Mechanism

- **First Stage:** Featuring 1.625-inch flex wheels operating at 400 RPM, the initial stage employs custom two-tooth gears. These gears replace traditional shaft inserts, preventing flex wheel deformation and slippage. The system is chain-driven for consistent power transmission.
- **Ramp Design:** The intake ramp hovers just above the field tiles at a 45-degree angle, facilitating a seamless transition for rings from the field to the conveyor system.



Conveyor System

- **Configuration:** The conveyor is synchronized with the intake via a chain drive and operates at 800 RPM, powered by 1.5 motors delivering 16.5 watts. This setup ensures rapid and reliable ring transport.
- **Top Flex Wheels:** Positioned at the conveyor's apex, these wheels provide necessary friction, exerting downward force to securely place rings onto stakes.
- **Hook Design:** Custom hooks, laser-cut from Delrin, are mounted using low-profile spacers. This design choice mitigates vibration-induced dislodgement, ensuring rings rest securely during transport.
- **Optical Sensors:** Integrated sensors detect ring colors, enabling the system to identify and reject incor rings, thereby enhancing scoring efficiency.



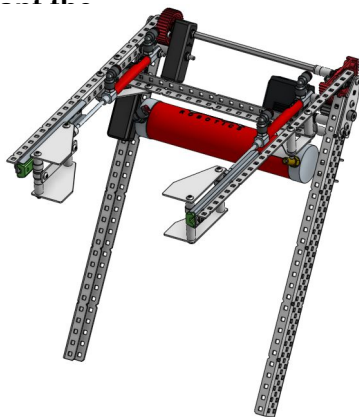
Doinker Mechanism

- **Goal Rush Assistance:** During the pre-game autonomous period, the doinker aids in swiftly securing mogos, giving the robot a competitive edge in goal acquisition.
- **Field Clearing:** Equipped with a Delrin section actuated by a piston, the doinker effectively clears field corners during autonomous routines and skill challenges.



Wall Stake Arm

- **Motor and Gear Ratio:** Operating with 0.5 motors (5.5 watts) and a 1:5 gear ratio, the arm achieves a controlled speed of 40 RPM, suitable for precise ring placement.
- **Piston Integration:** Dual pistons provide the arm a range of motion, enabling it to place wall stakes from various positions across the field.
- **Rotational Sensor:** Mounted on a joint with a brass insert, the sensor provides accurate positional feedback. This data supports macro programming for driver control, allowing single commands to position the arm precisely.
- **Joint Design:** Utilizing a screw joint at the arm's apex enhances stability and reduces play, ensuring consistent performance during ring placement.



Climbing Mechanism

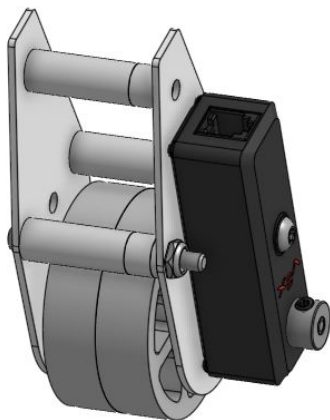
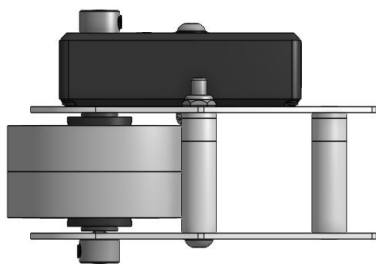
- **Tier 1 Climb:** The robot is engineered to ascend to the first tier, securing three points during the endgame. The wall stake arm, when fully extended, positions the climbing mechanism optimally for a successful hang.

Clamp System

- **Piston Configuration:** Dual 25mm pistons operate the clamp, providing robust and reliable grasping force.
- **Grip Enhancement:** The rear of the clamp features two rubber bumpers with a high coefficient of friction, ensuring a secure hold on mogos.
- **Alignment Aid:** A polycarbonate crossbar spans the drivetrain, assisting in precise mogo alignment during clamping.
- **Elastic Assistance:** Elastic bands apply downward force on the clamp, reducing pneumatic demand and conserving air supply.
- **Versatility:** The clamp's design allows it to secure mogos from multiple orientations, including corners and sides.
- **Ring Management:** Side-mounted polycarbonate pieces deflect rings, preventing entanglement with hooks and ensuring maximum ring capacity on stakes.

Odometry Wheels

- **Configuration:** The robot employs both vertical and angular odometry wheels, providing comprehensive tracking of movement.
- **Sensor Integration:** Rotational sensors relay precise positional data to the robot's brain, enabling accurate navigation and autonomous function execution.
- **Customization:** Laser-cut Delrin components ensure the odometry wheels are tailored to exact specifications, enhancing measurement accuracy.





Innovative Chapter



The Slip n' Slide Mech

Problem arm length:

When creating our wall stakes mechanism, we recognized the pros and cons of different arm lengths

Long Arm:

pros	cons
- Autonomous neutral wall stake - Can score if another robot is defending	- Loose fit

Short Arm:

pros	cons
- Aligner accuracy - Snug fit - 360 degrees	- Lack of distance for scoring on wall stakes

With our table we recognized how longer arms perform better in autonomous and shorter arms generally perform better for driver.

The Slip n' Slide Mech

By putting linear slides on the arm, it gives our arm two different lengths. This gives us the highlighted advantages while neglecting the disadvantages.

pros	cons
- Autonomous neutral wall stake - Can score if another robot is defending	- Loose fit
pros	cons
- Aligner accuracy - Snug fit - 360 degrees	- Lack of distance for scoring on wall stakes

The only pro we miss out on is the ability to allow our arm to turn 360 theoretical degrees while staying within the robot size limitations.

CAD Models Slip n' Slide

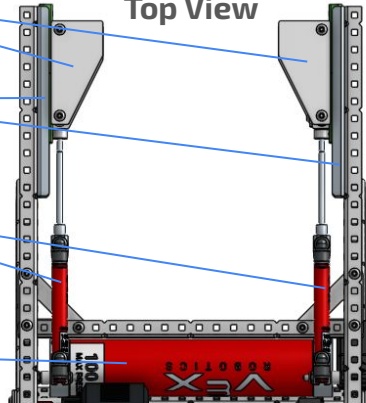
Delrin Basket

Linear Slide Track

Pneumatic Cylinders

Pneumatic Reservoir

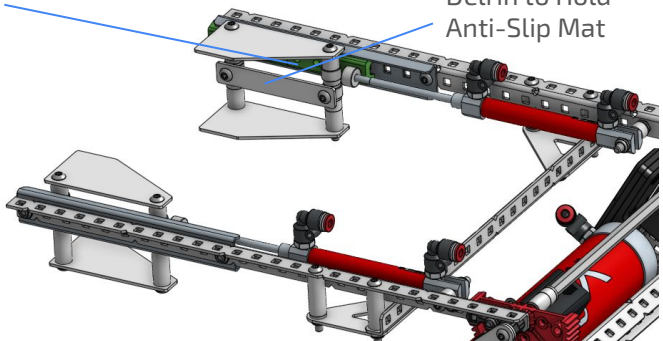
Top View



Dimetric View

Inner Slide Track

Delrin to Hold Anti-Slip Mat



Building Slip n' Slide

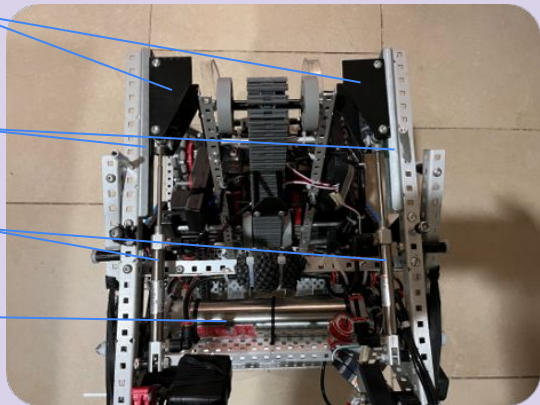
Top View

Delrin Basket

Linear Slide Track

Pneumatic Cylinders

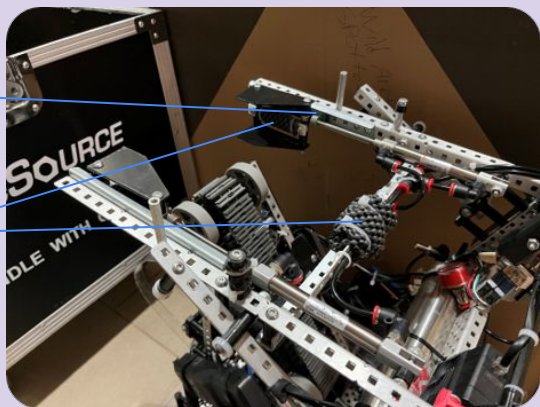
Pneumatic Reservoir



Dimetric View

Inner Slide Truck

Anti-Slip Mat



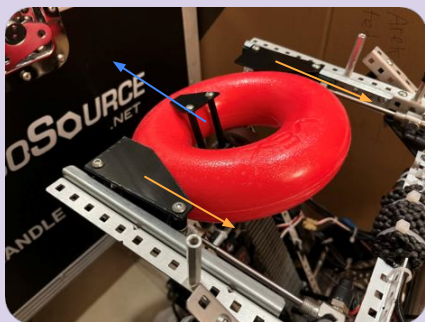
Problem Loading the Basket

After building the arm, we found it difficult to load the rings securely.

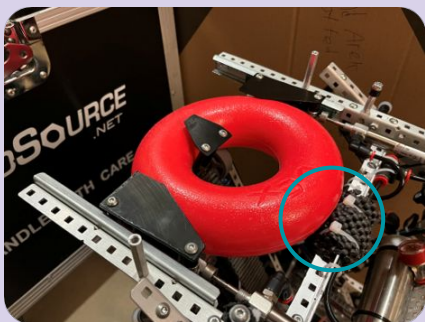
Solution Force of Pistons & Anti-Slip Mat

When loading the rings into the basket, it is loaded when the arm is in the long position. This meant that we could switch the arm to the short position to shove the ring deeper into the basket.

Before



After



↑ The force of the intake on the ring

↑ The force of the pistons on the ring

○ The point where the ring contacts anti-slip mat